EXPERIMENT 1: BATCH STERILIZATION

AIM:
To study microbial growth in a medium using the batch sterilization method.

PRINCIPLE:
Sterilization removes or destroys microorganisms from media and equipment. Moist heat sterilization is commonly performed using an autoclave under pressure. Sterile conditions help prevent contamination during microbial cultivation.

REQUIREMENTS:
Nutrient broth components, distilled water, conical flask, cotton, autoclave, pH meter, electronic balance, and spectrophotometer.

PROCEDURE:

1. Prepare nutrient broth using peptone, sodium chloride, beef extract, yeast extract, and distilled water.
2. Adjust the pH and plug the flask with cotton.
3. Sterilize the medium at 121°C and 15 psi for 15–20 minutes.
4. Transfer the sterile medium into the bioreactor and inoculate with E. coli culture.
5. Maintain suitable temperature and agitation during incubation.
6. Collect samples at regular intervals and measure optical density using a spectrophotometer at 325 nm.

OBSERVATION:
Optical density readings were recorded at regular time intervals. The changes in OD indicated the growth of E. coli in the medium.

RESULT:
The growth of E. coli was monitored by measuring optical density, and the bacterial growth phases were identified.

EXPERIMENT 2: ESTIMATION OF GROWTH KINETIC PARAMETERS IN BACTERIAL CELLS

AIM:
To estimate bacterial growth kinetic parameters by measuring optical density and plotting a growth curve.

PRINCIPLE:
Bacteria multiply by binary fission and show lag, exponential, stationary, and death phases. Optical density at 600 nm is used to estimate cell concentration. Growth data are used to calculate specific growth rate, generation time, and growth rate constant.

REQUIREMENTS:
Nutrient broth, E. coli culture, conical flask, incubator shaker, spectrophotometer, cuvettes, micropipettes, and distilled water.

PROCEDURE:

1. Prepare sterile nutrient broth and inoculate it with bacterial culture.
2. Incubate at 37°C with shaking at 150 rpm.
3. Collect culture samples at regular intervals.
4. Measure the optical density at 600 nm and record the readings.
5. Plot OD600 against time and identify the exponential phase.
6. Calculate the growth kinetic parameters.

OBSERVATION:
OD600 readings were recorded at different time intervals and used to plot the bacterial growth curve. The exponential phase was identified for calculating the growth parameters.

RESULT:
The bacterial growth curve was obtained, and specific growth rate, generation time, and growth rate constant were estimated.

CALCULATIONS:
Specific growth rate: $\mu = (\ln X_2 - \ln X_1) / (t_2 - t_1)$

Generation time: $g = 0.693 / \mu$

Growth rate constant: $k = 1 / g$

EXPERIMENT 3: DETERMINATION OF VOLUMETRIC OXYGEN TRANSFER COEFFICIENT (KLa)

AIM:
To determine the volumetric oxygen transfer coefficient (KLa) of a fermentation system using the dynamic gassing-out method.

PRINCIPLE:
Oxygen is essential for the growth of aerobic microorganisms. KLa represents the efficiency of oxygen transfer from air into the liquid medium. It is determined by measuring the increase in dissolved oxygen concentration with time.

REQUIREMENTS:
Bioreactor, dissolved oxygen probe, air compressor, nitrogen cylinder, sparger, impeller, medium or distilled water, and stopwatch.

PROCEDURE:

1. Fill the bioreactor with the required volume of medium or distilled water and calibrate the DO probe.
2. Sparge nitrogen until the dissolved oxygen level approaches zero.
3. Stop nitrogen flow and introduce air while maintaining agitation.
4. Record dissolved oxygen readings at regular intervals until saturation.
5. Calculate the oxygen deficit and plot ln(C* − CL) against time.
6. Determine KLa from the slope of the graph.

OBSERVATION:
Dissolved oxygen readings increased after air was introduced into the reactor. The recorded values were used to plot the oxygen transfer graph and calculate KLa.

RESULT:
The volumetric oxygen transfer coefficient of the fermentation system was determined using the dynamic gassing-out method.

CALCULATION:
KLa = − slope of the plot of ln(C* − CL) against time.

EXPERIMENT 4: DETERMINATION OF MIXING TIME IN A STIRRED TANK REACTOR

AIM:
To determine the mixing time of a stirred tank reactor using the tracer method.

PRINCIPLE:
Mixing ensures uniform distribution of nutrients, temperature, and dissolved oxygen in a bioreactor. Mixing time is the time required for a tracer to become uniformly distributed in the liquid. It can be determined by monitoring conductivity after adding a tracer.

REQUIREMENTS:
Stirred tank reactor, mechanical stirrer, impeller, baffles, stopwatch, conductivity meter, sodium chloride solution, and distilled water.

PROCEDURE:

1. Fill the stirred tank reactor with distilled water and set the agitation speed.
2. Add a small amount of sodium chloride solution as the tracer and start the stopwatch.
3. Record conductivity readings at regular intervals until a constant value is reached.
4. Determine the time required to reach approximately 95% of the final conductivity.
5. Repeat at different agitation speeds and compare the mixing times.

OBSERVATION:
Conductivity readings were recorded after adding the tracer at different agitation speeds. The time required to reach approximately 95% of the final conductivity was noted as the mixing time.

RESULT:
The mixing time was determined using the tracer method, and the effect of agitation speed on mixing efficiency was studied.

CALCULATION:
Mixing time (t_m) = $t_f - t_0$

Where, t_f = time at 95% mixing and t_0 = time of tracer addition.

EXPERIMENT 5: ESTIMATION OF CELL MAINTENANCE COEFFICIENT AND TRUE GROWTH YIELD

AIM:
To estimate the cell maintenance coefficient (ms) and true growth yield coefficient (YX/S,max) using mass and energy balance during cell growth.

PRINCIPLE: Microorganisms use substrates for both cell growth and maintenance activities. Pirt's equation describes the relationship between substrate consumption, biomass yield, and specific growth rate. A graph is used to estimate the maintenance coefficient and true growth yield.

REQUIREMENTS:
Bacterial culture, nutrient medium with glucose, fermenter or shake flask, spectrophotometer, glucose estimation reagent, incubator shaker, centrifuge, and pH meter.

PROCEDURE:

1. Prepare sterile nutrient medium containing a known glucose concentration and inoculate it with bacterial culture.
2. Incubate under suitable temperature and shaking conditions.
3. Collect samples at regular intervals and measure OD600.
4. Estimate the remaining glucose concentration using the DNS method or a suitable glucose assay.
5. Calculate biomass yield and specific growth rate.
6. Plot 1/YX/S against 1/μ to estimate the maintenance coefficient and true growth yield.

OBSERVATION: Biomass concentration and residual glucose were measured at different time intervals. These values were used to calculate biomass yield and estimate the maintenance coefficient and true growth yield.

RESULT: The cell maintenance coefficient and true growth yield coefficient were estimated using Pirt's equation.

CALCULATIONS:

Biomass yield coefficient: $Y_{X/S} = \Delta X / \Delta S$

Specific growth rate: $\mu = (\ln X_2 - \ln X_1) / (t_2 - t_1)$

Pirt's equation: $1/Y_{X/S} = 1/Y_{X/S,max} + m_s/\mu$

EXPERIMENT 6: DETERMINATION OF RESIDENCE TIME DISTRIBUTION (RTD) FOR A CSTR

AIM:

To determine the residence time distribution of a continuous stirred tank reactor using the pulse tracer method.

PRINCIPLE:

Residence time distribution describes the time fluid elements spend inside a reactor. A non-reactive tracer is introduced as a pulse, and its concentration is measured at the outlet over time. The concentration data are used to evaluate the reactor's flow behaviour.

REQUIREMENTS:

CSTR, feed reservoir, peristaltic pump, mechanical stirrer, conductivity meter, sodium chloride solution, stopwatch, beakers, and pipettes.

PROCEDURE:

1. Fill the CSTR with distilled water and maintain a constant flow rate and agitation.
2. Inject a small pulse of sodium chloride solution at the inlet and start the stopwatch.
3. Collect outlet samples at regular intervals.
4. Measure the conductivity of each sample until the tracer concentration approaches zero.
5. Record the readings and plot tracer concentration against time.
6. Calculate the RTD parameters and mean residence time.

OBSERVATION:

Tracer concentration was measured in outlet samples collected at regular time intervals. The concentration-time data were used to calculate the RTD and mean residence time.

RESULT: The residence time distribution of the CSTR was determined using the pulse tracer method.

CALCULATIONS:

Theoretical residence time: $\tau = V / Q$

Exit age distribution: $E(t) = C(t) / \int C(t)dt$

Mean residence time: $\tau = \Sigma(t_i C_i) / \Sigma C_i$

EXPERIMENT 7: STUDIES ON THE KINETICS OF IMMOBILIZED ENZYMES AND IMMOBILIZED CELLS

AIM:
To study the kinetics of immobilized enzymes and cells by determining reaction rates at different substrate concentrations.

PRINCIPLE:
Immobilization confines enzymes or cells within or onto a support while retaining their biological activity. It allows easy recovery, reuse, and improved stability of biocatalysts. Reaction rates at different substrate concentrations are used to estimate Vmax and Km.

REQUIREMENTS:
Immobilized enzyme beads, immobilized cell beads, substrate solutions, conical flasks, water bath, spectrophotometer, micropipettes, sodium alginate, calcium chloride, and DNS reagent.

PROCEDURE:

1. Prepare substrate solutions of different concentrations.
2. Add a fixed quantity of immobilized enzyme beads to each reaction flask.
3. Incubate at a suitable temperature for a fixed time.
4. Measure product formation using the DNS method or a suitable assay.
5. For immobilized cells, add the beads to fresh substrate solution and measure substrate utilization or product formation.
6. Calculate reaction rates and plot reaction rate against substrate concentration.
7. Estimate Vmax and Km using the Michaelis–Menten or Lineweaver–Burk plot.

OBSERVATION:
Product formation was measured at different substrate concentrations using immobilized enzymes or cells. The reaction rates were calculated and used to estimate the kinetic parameters.

RESULT:
The kinetic parameters Vmax and Km were estimated, and the catalytic behaviour of the immobilized system was studied.

CALCULATIONS:
Reaction rate: $v = \text{Product formed} / \text{Reaction time}$

Michaelis–Menten equation: $v = V_{max}[S] / (K_m + [S])$

Lineweaver–Burk equation: $1/v = (K_m/V_{max})(1/[S]) + 1/V_{max}$